

HERA-PdM: Uncertainty-Calibrated Escalation in a Hierarchical Edge-Digital-Twin Predictive Maintenance Framework—An Empirical Study on Bearing Run-to-Failure Data

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Abstract—Hierarchical predictive-maintenance (PdM) systems increasingly combine edge models, Digital Twins and cloud prognostics, yet the rule that decides when the edge must escalate is usually an anomaly threshold or an uncalibrated confidence score. We study HERA-PdM, a hierarchical edge-Digital-Twin framework in which conformalized quantile regression (CQR) produces remaining-useful-life (RUL) intervals, the edge escalates only when its interval does not determine a unique maintenance action (which bounds unsafe edge decisions by the miscoverage level), and high-impact actions are issued as SHAP-supported recommendations for human approval. On all 17 FEMTO/PRONOSTIA bearings with bearing-grouped cross-validation and three seeds, a 2.5k-parameter edge GRU was the most accurate model (RMSE 816s versus at least 1047s for fog/cloud models), and CQR raised interval coverage from 0.31–0.69 to 0.83–0.89 (nominal 0.90). Point estimates with a fixed threshold left 41.0% of critical snapshots without maintenance action, versus 5.4–9.4% for calibrated policies, at the price of 75.0–86.8% false maintenance. However, the escalation gate sent 87.5% of snapshots upward and was not better than width-, score- or random-based gates at matched budget. The results locate the bottleneck of hierarchical PdM on this benchmark in failure predictability rather than in inference routing.

Index Terms—predictive maintenance, remaining useful life, conformal prediction, edge computing, Digital Twin, human-in-the-loop, Industry 5.0

I. INTRODUCTION

Predictive maintenance (PdM) estimates the remaining useful life (RUL) of machinery from condition-monitoring data so that interventions can be scheduled before failure [1]. Deep sequence models are the prevailing RUL estimators [2], [3], but they are usually trained and evaluated as centralized point predictors. In a factory, three further requirements appear. First, vibration streams are produced at the machine, so edge-cloud architectures that keep routine inference local have been proposed [4]. Second, a point estimate says nothing about how far it can be trusted; RUL uncertainty must therefore be quantified and, ideally, calibrated [5], [6]. Third, Industry 5.0 places the human at the centre of such systems [7]: decision-makers often do not adopt system-generated maintenance advice [8], so high-impact actions should remain explained recommendations.

Recent systems integrate these elements with agents and Digital Twins (DTs) [9]–[12]. Our literature audit (42 papers;

Section II) shows that the *combination* of agents, DTs, edge inference and human oversight is no longer new. What remains unclear is the *escalation rule*: current hierarchical PdM systems escalate on anomaly thresholds [10] or on uncalibrated confidence [13], whereas calibrated conformal deferral has so far been studied for classification cascades [14]. We therefore ask whether conformally calibrated RUL intervals computed on the edge can serve as the escalation signal of a hierarchical PdM system, and at what cost in accuracy, safety, communication and latency.

The contributions are as follows:

- 1) **HERA-PdM**, a reproducible four-layer framework (edge monitoring, synchronized DT state, CQR prognostics, human-approved decisions) with a *decision-sufficiency* gate that escalates only when the edge RUL interval straddles a maintenance-decision boundary; this bounds locally resolved unsafe decisions by the edge miscoverage level.
- 2) A bearing-grouped evaluation on all 17 FEMTO bearings of ten quantile models (with and without DT context) and point baselines, showing that the smallest (edge) model was the most accurate and that CQR restored, but did not fully reach, nominal coverage.
- 3) Decision-level evidence that calibration prevents most unsafe point-threshold decisions but that calibrated and margin-adjusted point policies share one safety/false-alarm frontier.
- 4) A system-level negative result: at the nominal level the gate escalated most snapshots and offered no advantage over simpler gates, which identifies failure predictability as the limiting factor.

II. RELATED WORK AND RESEARCH GAP

Deep RUL prediction. LSTM [2], multiscale CNN for bearings [3] and TCN [15] models are standard RUL estimators; reviews [1] treat health-stage division, i.e. detecting when degradation becomes observable, as a step preceding RUL prediction, which matters for the FEMTO bearings used here [16]. **Digital Twins and agents.** DTs provide a synchronized virtual state of an asset [9]. Agentic DT architectures for PdM, including human-centred ones, have appeared recently [11],

TABLE I
CLOSEST RELATED WORK (CODED FROM FULL TEXT, ABSTRACTS OR PUBLIC CODE; SEE LITERATURE MATRIX). ✓: PRESENT, ○: PARTIAL, -: ABSENT.

Work	DT	Edge	Agents	RUL	Calib.	UQ	Routing	XAI	HIL
Ren et al. [4]	-	✓	-	✓	-	-	-	-	-
HAMA [10]	-	✓	✓	○	-	-	○ anomaly	✓	○
EC-Distill [13]	-	✓	✓	-	-	-	✓ conf.	○	-
CAB cascade [14]	-	✓	-	-	✓	-	✓	-	-
Javanmardi [6]	-	-	-	✓	✓	-	-	-	-
Wang et al. [20]	-	-	-	✓	✓	-	-	-	-
Xu & Zhang [21]	-	-	○	✓	○	-	-	-	-
HERA-PdM	✓	✓	✓	✓	✓	-	✓	✓	✓

[12]; the survey in [12] lists hierarchical DT orchestration with residual-life estimation and explainability as an open question. *Edge and hierarchical PdM.* Ren et al. [4] run a lightweight TCN at the edge and refine RUL in the cloud; HAMA [10] escalates from a z-score edge pre-filter to fog ensembles with SHAP and generated operator text; EC-Distill-ZeroDiag [13] offloads uncertain fault-diagnosis cases from an edge agent to a cloud model. *Uncertainty quantification.* Bayesian, dropout, ensemble and heteroscedastic RUL uncertainty methods have been benchmarked in [5]; split conformal and CQR intervals give finite-sample coverage [17], [18] and have been applied to C-MAPSS [6], [19], where weighted variants were recommended for non-exchangeable conditions [19], and to bearings [20]. Probabilistic RUL has also been fed into risk-aware maintenance policies [21]. For inference cascades, confidence-based deferral [22] and conformal edge–cloud offloading for classification and numerical solvers [14], [23] provide the closest methodological precedents. *Human-centric PdM.* Behavioural studies of PdM adoption [8] motivate advisory, explained recommendations.

Gap. Table I summarizes the closest works. To the best of our literature review, no study uses conformal RUL intervals computed on the edge as the escalation rule of a hierarchical PdM system, relates escalation to maintenance-decision boundaries, and evaluates it against the gating rules of existing systems on public run-to-failure data with communication and latency accounting. The contribution is methodological and incremental; the architecture itself is not new.

III. HERA-PdM METHODOLOGY

Fig. 1 shows the four layers. The “agents” are software components with fixed responsibilities and message interfaces; no language model is used anywhere in the pipeline.

Edge Monitoring Agent. Every 10 s a 0.1 s two-axis vibration snapshot is reduced to $x_t \in \mathbb{R}^d$, $d=28$ (per axis: RMS, kurtosis, skewness, peak-to-peak, crest, impulse and shape factors, spectral centroid and six log band energies). Each feature is z-scored against the asset’s first 50 snapshots and passed through $\text{sign}(z) \log(1+|z|)$. The health index is $h_t = \log(1 + \frac{1}{d} \sum_j |z_{t,j}|)$, and the anomaly score a_t is the exponentially smoothed h_t standardized by the mean and spread of the asset’s raw baseline h_t ; the alarm threshold

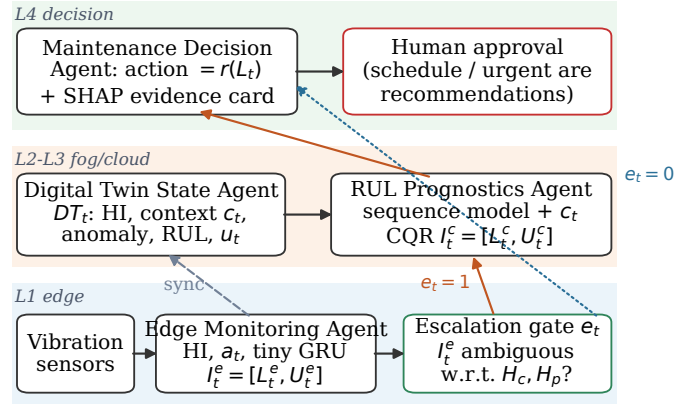


Fig. 1. HERA-PdM. The edge agent computes a calibrated RUL interval I_t^e ; the gate escalates only if I_t^e is not decision-sufficient. The fog/cloud prognostics agent uses the Digital Twin context c_t . Schedule/urgent actions require human approval.

τ_a is the 99th percentile of a_t on healthy training snapshots (17.0–18.4 across folds). A tiny model maps the last $W_e=16$ snapshots to three quantiles of the RUL.

Digital Twin State Agent. The twin is a synchronized state record $DT_t = \{\text{operating condition, sensor state, } h_t, \text{ degradation context } c_t, \hat{R}_t, I_t, u_t, \text{ anomaly state}\}$, updated causally from a 12-byte sync message (t, h_t, a_t) per snapshot. The context $c_t \in \mathbb{R}^6$ contains fast and slow exponential averages of h_t , their difference, the running maximum, a 30-snapshot slope and the cumulative excess of h_t over its baseline. Absolute operating time is deliberately excluded because bearing lifetimes range from 2.3×10^3 to 2.8×10^4 s.

RUL Prognostics Agent. For a window X_t of $W_c=64$ snapshots and context c_t , a sequence model g_ϕ outputs non-crossing quantiles $(\hat{q}_t^{\alpha/2}, \hat{q}_t^{0.5}, \hat{q}_t^{1-\alpha/2})$, with the outer quantiles parametrized as the median minus/plus a softplus term, trained with the pinball loss on the capped label $y_t = \min(\text{RUL}_t, R_{\max})$, $R_{\max}=3000$ s. The median $\hat{R}_t = \hat{q}_t^{0.5}$ is the point estimate.

Calibration. Split CQR [17] uses n snapshots of held-out calibration bearings with scores $s_i = \max(\hat{q}_i^{\alpha/2} - y_i, y_i - \hat{q}_i^{1-\alpha/2})$ and the $\lceil (n+1)(1-\alpha) \rceil$ -th smallest score $\hat{Q}_{1-\alpha}$:

$$I_t = [L_t, U_t] = [\hat{q}_t^{\alpha/2} - \hat{Q}_{1-\alpha}, \hat{q}_t^{1-\alpha/2} + \hat{Q}_{1-\alpha}] \cap [0, R_{\max}]. \quad (1)$$

If calibration and test assets are exchangeable, $\Pr(y_t \notin I_t) \leq \alpha$. Snapshots within a bearing are dependent, so this guarantee is only approximate here, which we test empirically. The width $u_t = U_t - L_t$ is the uncertainty.

Maintenance decision. With critical and planning horizons $H_c=600$ s and $H_p=1800$ s, the action region of a value v is $r(v) = \text{URGENT}$ if $v \leq H_c$, SCHEDULE if $H_c < v \leq H_p$ and CONTINUE otherwise. The Maintenance Decision Agent acts on the conservative bound, $\pi_t = r(L_t)$, and downgrades CONTINUE to INSPECT when $a_t > \tau_a$. SCHEDULE and URGENT are issued as recommendations with an explanation card (RUL, interval, anomaly evidence, top-3 SHAP [24])

contributors, rationale) and require human approval.

Decision-sufficiency escalation. Let $B = \{H_c, H_p\}$. The edge escalates when its interval is ambiguous with respect to B , or when anomaly evidence contradicts a healthy verdict:

$$e_t = \mathbb{I}[\exists h \in B: L_t^e \leq h < U_t^e] \vee \mathbb{I}[a_t > \tau_a \wedge r(L_t^e) = C], \quad (2)$$

where C denotes CONTINUE. If $e_t = 0$, the edge action $r(L_t^e)$ is final; otherwise the cloud action $r(L_t^c)$ is used. A local CONTINUE requires $L_t^e > H_p \geq H_c$, so a critical asset ($y_t \leq H_c$) can only be left running locally if $y_t < L_t^e$; hence

$$\Pr(e_t=0, \pi_t=\text{CONTINUE}, y_t \leq H_c) \leq \Pr(y_t \notin I_t^e) \leq \alpha_e, \quad (3)$$

and the same argument bounds locally issued false URGENT actions. The edge miscoverage α_e is thus the single safety knob of the gate. Unlike an interval-width threshold, the gate ignores wide intervals that lie inside one action region and escalates narrow ones that cross a boundary.

Communication and compute model. Payloads are counted analytically in float32: always-cloud streams x_t (112 B/snapshot); HERA sends the 12 B sync every snapshot plus the untransmitted part of the W_c window when $e_t=1$. Decision latency is edge time + $e_t(\text{RTT}+\text{cloud time})$ with an assumed RTT of 50 ms.

IV. EXPERIMENTAL SETUP

Data. FEMTO-ST/PRONOSTIA (IEEE PHM 2012) [16]: 17 bearings run to failure under three conditions (1800 rpm/4000 N, 7 bearings; 1650 rpm/4200 N, 7; 1500 rpm/5000 N, 3), two accelerometers at 25.6 kHz, 24,889 snapshots. The full test trajectories (Full_Test_Set) were used, so every bearing has a known failure time.

Protocol. Bearings are split into four condition-balanced folds; each bearing is tested exactly once. In every fold, one bearing per condition is held out for conformal calibration and the rest are used for training. Hyper-parameters were selected on inner validation bearings drawn from training bearings only: AdamW, one-cycle learning rate 2×10^{-3} , weight decay 10^{-3} , 8 epochs, and for fog/cloud models width 32 and input noise 0.2 (edge models: width 16, no noise). Architectures were chosen on calibration-set pinball loss, giving Edge-GRU (edge) and CNN-LSTM+DT (fog/cloud). Every model was trained with seeds 0–2; metrics are pooled over the 17 test bearings and reported as mean $\pm$ std over seeds. The paired Wilcoxon signed-rank test uses bearings as units.

Baselines. Prognostics: LSTM, GRU, TCN [15], CNN-LSTM, each with and without DT context, plus MSE-trained point models. Policies: always-edge, always-cloud, cloud point estimate with fixed thresholds, and ablations removing uncertainty (point estimate with a margin equal to the edge calibration RMSE), the DT context, or the anomaly term, or replacing the gate by an anomaly trigger (as in [10]), by interval width (confidence deferral [22]), by a linear score of anomaly, width and risk, or by random escalation; the last three use the HERA escalation budget measured on calibration bearings.

TABLE II
RUL ACCURACY AND CALIBRATION ON 17 HELD-OUT FEMTO BEARINGS (4 BEARING-GROUPED FOLDS; MEAN $\pm$ STD OVER 3 SEEDS; RUL CAPPED AT 3000 s; NOMINAL COVERAGE 90%). RAW: UNCALIBRATED QUANTILES; CQR: CONFORMALIZED. [†]SELECTED ON CALIBRATION BEARINGS.

Model	Params	RMSE (s)	MAE (s)	Raw cov.	CQR cov.	MPIW (s)	CCE
Edge-CNN	2.7k	1114 $\pm$ 40	851 $\pm$ 19	0.37	0.88 $\pm$ 0.02	2080 $\pm$ 120	0.035
Edge-GRU [†]	2.5k	816 $\pm$ 30	525 $\pm$ 65	0.69	0.84 $\pm$ 0.04	1879 $\pm$ 130	0.054
LSTM	17.5k	1200 $\pm$ 79	778 $\pm$ 69	0.53	0.86 $\pm$ 0.04	2014 $\pm$ 184	0.072
LSTM+DT	17.7k	1202 $\pm$ 127	824 $\pm$ 119	0.44	0.86 $\pm$ 0.01	1989 $\pm$ 28	0.066
GRU	13.4k	1149 $\pm$ 95	706 $\pm$ 87	0.54	0.85 $\pm$ 0.03	2009 $\pm$ 167	0.067
GRU+DT	13.6k	1166 $\pm$ 83	784 $\pm$ 77	0.38	0.88 $\pm$ 0.04	2145 $\pm$ 244	0.051
TCN	16.8k	1112 $\pm$ 63	825 $\pm$ 79	0.32	0.87 $\pm$ 0.02	2122 $\pm$ 170	0.032
TCN+DT	17.2k	1047 $\pm$ 77	808 $\pm$ 49	0.31	0.89 $\pm$ 0.02	2314 $\pm$ 111	0.040
CNN-LSTM	15.3k	1096 $\pm$ 104	676 $\pm$ 105	0.47	0.83 $\pm$ 0.03	2147 $\pm$ 104	0.044
CNN-LSTM+DT [†]	15.7k	1093 $\pm$ 64	717 $\pm$ 43	0.36	0.85 $\pm$ 0.01	2008 $\pm$ 75	0.062
LSTM+DT (MSE)	–	1288 $\pm$ 64	945 $\pm$ 76	–	–	–	–
GRU+DT (MSE)	–	1298 $\pm$ 88	959 $\pm$ 117	–	–	–	–
TCN+DT (MSE)	–	1114 $\pm$ 26	879 $\pm$ 25	–	–	–	–
CNN-LSTM+DT (MSE)	–	1100 $\pm$ 66	789 $\pm$ 45	–	–	–	–

Metrics. RMSE and MAE (s); coverage (PICP), mean interval width (MPIW) and coverage calibration error (CCE, mean |nominal–empirical| over $\alpha \in [0.05, 0.5]$); detection of $y_t \leq H_c$ (precision, recall, F1, AUROC); *unsafe rate* (critical snapshots with CONTINUE/INSPECT); *false maintenance* (FM: SCHEDULE/URGENT when $y_t > H_p$); escalation rate, payload, MACs and latency. The NASA scoring function was not used because it is defined for C-MAPSS cycles. Latency is single-thread CPU wall-clock time on an Apple M2 Pro host, not on embedded hardware; no energy was measured.

V. RESULTS AND DISCUSSION

RQ1: accuracy. Table II shows that the 2.5k-parameter Edge-GRU (34k MACs) achieved the lowest error (RMSE 816 s, MAE 525 s), whereas the best fog/cloud model reached 1047 s (TCN+DT). DT context did not help consistently: it lowered RMSE for TCN but not for GRU or LSTM. Larger windows and capacity did not transfer to unseen bearings, so escalation cannot be justified by accuracy on this benchmark. Quantile training did not hurt point accuracy relative to MSE training.

RQ2: calibration. Raw quantiles covered only 0.31–0.69 of test snapshots; CQR raised coverage to 0.83–0.89 (CCE ≤ 0.08) with wide intervals (MPIW 1879 s for the edge). Coverage stayed below 0.90 and varied strongly across bearings: 11 of 17 bearings fell below 0.90 under HERA’s final intervals, and the lowest per-bearing edge coverage was 0.13 (seed mean), mostly on the 230-snapshot Bearing 2_7. With three calibration bearings per fold, exchangeability between bearings is the binding assumption; weighted conformal methods [19] are a natural remedy. Fig. 2 shows typical behaviour: intervals narrow when degradation becomes visible and stay wide during the long healthy phase. An asymmetric CQR variant that calibrates each bound at $\alpha/2$ gave similar coverage (0.83) and width.

RQ4: unsafe decisions. Table III shows that the cloud point estimate with fixed thresholds left 41.0% of critical snapshots without maintenance action, whereas calibrated policies left 5.4% (always-cloud) to 9.4% (HERA). Across bearings the

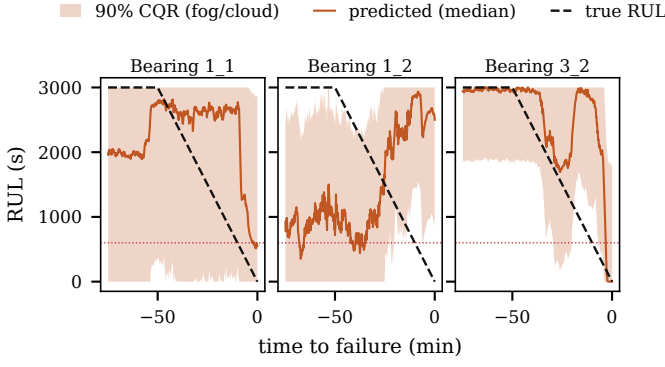


Fig. 2. Fog/cloud (CNN-LSTM+DT) median RUL and 90% CQR interval in the last 75 min of three test bearings (seed 0; bearings with the most degradation-phase snapshots). Dotted line: H_c .

TABLE III

HIERARCHICAL POLICIES, ABLATIONS AND SYSTEM COST (ALL 17 BEARINGS, MEAN $\pm$ STD OVER 3 SEEDS). ESC.: ESCALATED SNAPSHOTS; UNSAFE: CRITICAL SNAPSHOTS ($y \leq H_c$) LEFT WITHOUT MAINTENANCE ACTION; FM: FALSE MAINTENANCE ON HEALTHY SNAPSHOTS ($y > H_p$); B/S: PAYLOAD BYTES PER SNAPSHOT; LAT.: MEAN DECISION LATENCY WITH AN ASSUMED 50 MS RTT. [†]ESCALATION BUDGET MATCHED TO HERA ON CALIBRATION BEARINGS.

Policy	Esc. (%)	Unsafe (%)	FM (%)	RMSE (s)	B/s	Lat. (ms)
Always-edge	0.0	6.7 $\pm$ 5.9	86.8 $\pm$ 15.4	809	0	0.1
Always-cloud	100.0	5.4 $\pm$ 1.3	84.0 $\pm$ 5.6	1090	112	50.2
Cloud point + fixed thr.	100.0	41.0 $\pm$ 3.4	23.6 $\pm$ 1.4	1100	112	50.2
HERA-full	87.5	9.4 $\pm$ 3.7	75.0 $\pm$ 11.6	1083	117	44.1
HERA w/o anomaly term	87.4	9.6 $\pm$ 3.8	75.0 $\pm$ 11.6	1084	117	44.0
HERA-no-DT	87.5	6.5 $\pm$ 5.7	82.6 $\pm$ 11.2	1088	117	44.1
HERA-no-UQ (point margin)	32.7	48.2 $\pm$ 5.5	17.0 $\pm$ 4.4	1014	69	16.5
HERA-fixed (anomaly trig.)	3.9	6.7 $\pm$ 5.7	86.8 $\pm$ 15.4	811	27	2.1
Width gate [†]	87.4	9.5 $\pm$ 3.8	75.0 $\pm$ 11.6	1084	117	44.0
Linear score gate [†]	88.9	7.6 $\pm$ 2.1	77.0 $\pm$ 10.6	1086	117	44.8
Random gate [†]	85.2	6.1 $\pm$ 0.9	81.8 $\pm$ 4.7	1059	124	42.9

reduction relative to the point policy was significant ($p < 0.001$, 14 non-zero pairs), and so was the increase in false maintenance ($p < 0.001$). Fig. 3(a) sweeps α for calibrated policies and a safety margin δ for point policies: all policies fall on one safety/false-alarm frontier. Calibration therefore helps by choosing an operating point from held-out data with a stated (though here only approximately met) guarantee, rather than by producing better decisions than a well-tuned margin. The remaining unsafe decisions concentrate on Bearings 2_3 and 2_7, whose vibration features show no precursor before failure.

RQ3 and RQ5: escalation and system trade-off. Because 90% intervals were about 1879 s wide on a 3000 s range, most edge intervals straddled H_p (Fig. 4), and HERA escalated 87.5% of snapshots (Fig. 3(b)). At this budget the width gate behaved almost identically (9.5% unsafe, 75.0% FM), and the linear-score and random gates were within a few points; differences across bearings were not testable (2 non-zero pairs). Relaxing the edge level to $\alpha_e = 0.3$ reduced escalation to 74.1% but raised unsafe decisions to 14.1%, consistent with the bound α_e . The anomaly-trigger gate escalated only 3.9% of snapshots because the anomaly score detected few critical

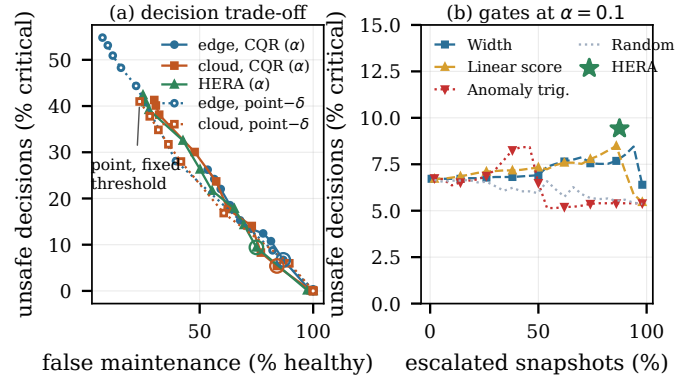


Fig. 3. (a) Unsafe decisions vs. false maintenance when sweeping α (calibrated) or margin δ (point); circles mark $\alpha = 0.1$. (b) Unsafe decisions vs. escalation budget for gates operating on the same $\alpha = 0.1$ edge intervals.

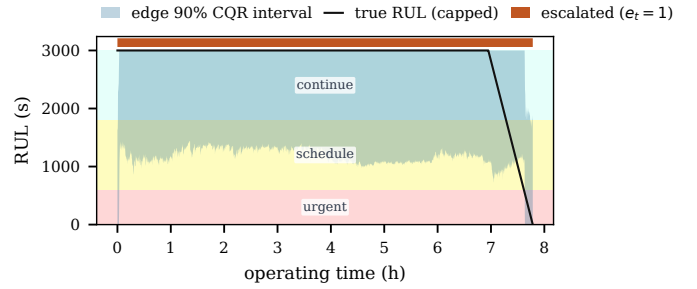


Fig. 4. Decision-sufficiency gate on test Bearing 1_1 (seed 0). Shaded bands: action regions. During the healthy phase the edge interval straddles H_p , so the gate escalates almost continuously (top bar).

snapshots (AUROC 0.81, F1 0.27; Isolation Forest AUROC 0.77). HERA’s payload (116.6 B/snapshot) slightly exceeded always-cloud feature streaming (112.0 B), and its mean latency (44.1 ms) was dominated by the assumed RTT; the edge model alone needed 0.13 ms and the fog model 0.24 ms per inference. Removing the DT context (HERA-no-DT) or the anomaly term changed the unsafe/FM operating point but not the escalation rate.

Explainability and human oversight. Expected-gradient SHAP on 300 degradation-phase snapshots of one fold (Fig. 5) ranked the DT feature “cumulative excess health index”, the horizontal spectral centroid and the 0–1 kHz band energy highest; the attributions were approximate (mean additivity error 154.8 s). An automatically generated card for the most critical explained snapshot reads:

Asset Bearing1_3, snapshot 2078. **RUL** 434 s, 90% interval [0, 2194] s. **Anomaly** $a_t = 15.8$ (threshold 17.04, no alarm). **Top SHAP** h_{centroid} (+691 s), $h_{\text{band0_1k}}$ (-508 s), v_{skew} (-369 s). **Recommendation** urgent maintenance — requires human approval. **Rationale** lower bound 0 s $\leq H_c$.

All fields come from the predictors; the text is a fixed template, not generated by a language model. Because of wide intervals, 76.7% of HERA snapshots produced approval requests; this workload would overwhelm operators and shows that calibrated but uninformative intervals shift burden to humans rather than removing it.

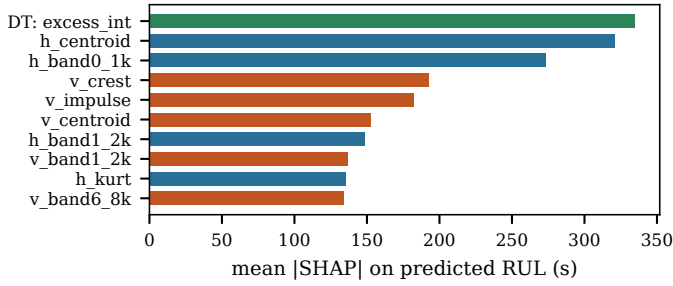


Fig. 5. Top-10 mean |SHAP| attributions of the fog/cloud median RUL (fold 2, seed 0). Green: DT context; blue/orange: horizontal/vertical vibration features.

Limitations. One dataset with 17 bearings and abrupt failures; approximate exchangeability; a lightweight state-based rather than physics-based DT; latency measured on a host CPU and communication counted analytically; no industrial deployment and no user study of the explanation cards.

VI. CONCLUSION

HERA-PdM uses calibrated RUL intervals to decide both when an edge device escalates and which maintenance action is recommended, and its gate bounds unsafe edge decisions by the edge miscoverage level. On FEMTO bearings, calibration prevented most unsafe decisions made by point-threshold policies, but the tiny edge model was the most accurate predictor, intervals were too wide for decision-sufficient local resolution, and the gate did not outperform simpler gates or reduce communication. The results suggest that, before routing rules are refined, hierarchical PdM needs features or sensing with longer warning lead times and calibration methods that account for between-asset heterogeneity; evaluating the gate on datasets with gradual degradation is the next step.

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